

ADVANCE JAVA

ASSIGNMENT - 03

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ROLL NO: 3098

Q1)

File name: error.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>

  <head>

    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>error Page</title>

  </head>

  <body>

    <h3 style="color:red;">Invalid username or password!</h3>
    <a href="login.jsp">Try Again</a>

  </body>

</html>
```

File name: login.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
```

```
<html>

<head>

  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">

  <title>login Page</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      background-color: orange;

      display: flex;

      justify-content: center;

      align-items: center;

      height: 100vh;

      margin: 0;

    }

    .login-container {

      background-color: green;

      padding: 30px 40px;

      border-radius: 10px;

      box-shadow: 0 4px 12px rgba(0, 0, 0, 0.1);

      text-align: center;

    }

    h2 {

      margin-bottom: 400px;

      color: #333;

    }

    input[type="text"], input[type="password"] {
```

```
width: 100%;  
padding: 10px 12px;  
margin: 10px 0;  
border: 1px solid #ccc;  
border-radius: 5px;  
font-size: 16px;  
}
```

```
input[type="submit"] {  
    background-color: #007bff;  
    color: white;  
    padding: 10px 20px;  
    border: none;  
    border-radius: 5px;  
    font-size: 16px;  
    cursor: pointer;  
}
```

```
input[type="submit"]:hover {  
    background-color: #0056b3;  
}
```

```
</style>
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>Login</h2>
```

```
<form method="post" action="login">
```

```
Username: <input type="text" name="username" required /><br><br>
```

```
Password: <input type="password" name="password" required /><br><br>
```

```
        <input type="submit" value="Login" />
    </form>
</body>
</html>
```

File name: welcome.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%
    String username = (String) request.getAttribute("username");
%>
<!DOCTYPE html>
<html>
    <head>
        <title>welcome</title>
    </head>
    <body>
        <h2>Welcome, <%= username %>!</h2>
    </body>
</html>
```

File name: login.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
```

```
/**
```

```
 * * @author Uday
```

```
 */
```

```
//  Servlet mapping using annotation
```

```
//@WebServlet("/login")
```

```
public class login extends HttpServlet {
```

```
    protected void processRequest( HttpServletRequest request, HttpServletResponse  
    response)
```

```
        throws ServletException, IOException {
```

```
        response.setContentType("text/html;charset=UTF-8");
```

```
        String username = request.getParameter("username");
```

```
        String password = request.getParameter("password");
```

```
        try (PrintWriter out = response.getWriter()) {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            java.sql.Connection con = java.sql.DriverManager.getConnection(  
                "jdbc:mysql://localhost:3306/ajava?zeroDateTimeBehavior=CONVERT_TO_NULL",  
                "root", "root"
```

```
            );
```

```
            java.sql.PreparedStatement ps = con.prepareStatement(  
                "SELECT * FROM users WHERE username = ? AND password = ?"
```

```
            );
```

```
            ps.setString(1, username);
```

```
            ps.setString(2, password);
```

```

java.sql.ResultSet rs = ps.executeQuery();

if (rs.next()) {
    // ✅ Forward to welcome.jsp with username
    request.setAttribute("username", username);
    request.getRequestDispatcher("welcome.jsp").forward(request, response);
} else {
    // Login failed
    response.sendRedirect("error.jsp");
}

con.close();
} catch (Exception e) {
    e.printStackTrace();
}
}

```

@Override

```

protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}

```

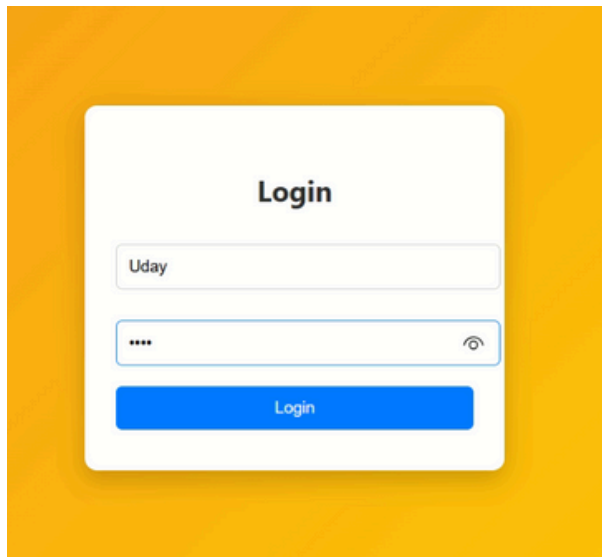
@Override

```

protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}

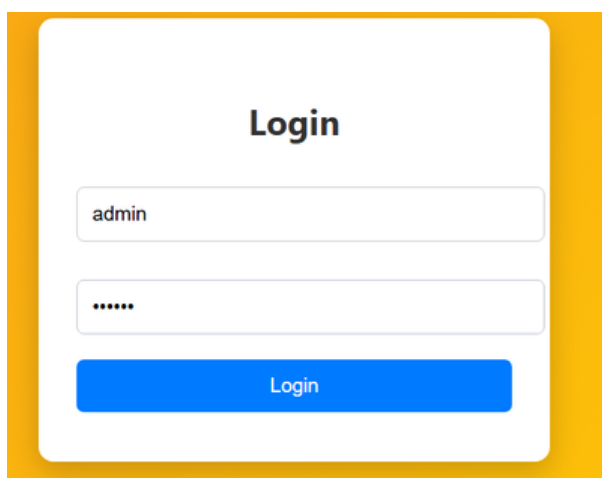
```

```
@Override  
public String getServletInfo() {  
    return "Login Servlet";  
}  
}
```



Invalid username or password!

[Try Again](#)



Welcome, admin! Login Successful....

Q2)

File name: display.jsp

```
<%@ page import="java.sql.*" %>  
<%@ page contentType="text/html;charset=UTF-8" language="java" %>  
<html>  
<head>  
    <title>Registered Users</title>
```

```

</head>
<body>
    <h2><%= request.getAttribute("message") %></h2>
    <h3>List of Registered Users:</h3>
    <table border="1">
        <tr>
            <th>ID</th>
            <th>First Name</th>
            <th>Last Name</th>
            <th>City</th>
            <th>Mobile</th>
            <th>Email</th>
        </tr>

        <%
            try {
                Class.forName("com.mysql.jdbc.Driver");

                Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/ajava?zeroDateTimeBehavior=C
ONVERT_TO_NULL", "root", "root");

                Statement stmt = con.createStatement();

                ResultSet rs = stmt.executeQuery("SELECT * FROM registration");

                while (rs.next()) {
                    %>
                    <tr>
                        <td><%= rs.getInt("id") %></td>
                        <td><%= rs.getString("first_name") %></td>
                        <td><%= rs.getString("last_name") %></td>
                        <td><%= rs.getString("city") %></td>

```



```

        <td><%= rs.getString("mobile") %></td>

        <td><%= rs.getString("email") %></td>

    </tr>

<%
    }

    con.close();

} catch (Exception e) {

    out.println("Error: " + e.getMessage());

}

%>

</table>

</body>

</html>

```

File name: registration.html

```

<html>
<head>
<title>registration form</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
<h2>Register New User</h2>
<form action="registration" method="post">
    ID: <input type="number" name="id" required><br><br>
    First Name: <input type="text" name="firstName" required><br><br>
    Last Name: <input type="text" name="lastName" required><br><br>
    City: <input type="text" name="city" required><br><br>
    Mobile: <input type="text" name="mobile" required><br><br>
    Email: <input type="email" name="email" required><br><br>
    <input type="submit" value="click for registration"/>
</form>
</body>
</html>

```

File name: registration.java

```

import jakarta.servlet.RequestDispatcher;

import jakarta.servlet.ServletException;

import jakarta.servlet.http.HttpServlet;

```

```

import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.*;
import java.sql.*;

public class registration extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        String firstName = request.getParameter("firstName");
        String lastName = request.getParameter("lastName");
        String city = request.getParameter("city");
        String mobile = request.getParameter("mobile");
        String email = request.getParameter("email");

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con =
                DriverManager.getConnection("jdbc:mysql://localhost:3306/ajava?zeroDateTimeBehavior=C
                ONVERT_TO_NULL", "root", "root");

            PreparedStatement ps = con.prepareStatement("INSERT INTO registration VALUES (?,
            ?, ?, ?, ?, ?)");

            ps.setInt(1, id);
            ps.setString(2, firstName);
            ps.setString(3, lastName);
            ps.setString(4, city);
            ps.setString(5, mobile);
            ps.setString(6, email);

```

```

        ps.executeUpdate();

        request.setAttribute("message", "User Registered Successfully!");
        RequestDispatcher rd = request.getRequestDispatcher("display.jsp");
        rd.forward(request, response);

    } catch (Exception e) {
        e.printStackTrace();
        response.getWriter().println("Error: " + e.getMessage());
    }
}
}

```

Register New User

ID:

First Name:

Last Name:

City:

Mobile:

Email:

User Registered Successfully!

List of Registered Users

ID	First Name	Last Name	City	Mobile	Email
3098	Uday	bholane	Ahmedabad	8956464646	uday23@gmail.com

Q3)

File name: Index.jsp

```
<%@ page contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>

<%@ page import="java.sql.*" %>

<html>

<head><title>Students List</title></head>

<body>

    <h2>All Students</h2>

    <a href="insert.jsp">Add New Student</a><br><br>

    <table border="1">

        <tr><th>ID</th><th>Name</th><th>Email</th><th>City</th><th>Actions</th></tr>

        <%

            try {

                Class.forName("com.mysql.cj.jdbc.Driver");

                Connection con =

                DriverManager.getConnection("jdbc:mysql://localhost:3306/ajava?zeroDateTimeBehavior=C

                ONVERT_TO_NULL", "root", "root");

                Statement stmt = con.createStatement();

                ResultSet rs = stmt.executeQuery("SELECT * FROM asg3_q3");

                while (rs.next()) {

                    %>

                    <tr>

                        <td><%= rs.getInt("id") %></td>

                        <td><%= rs.getString("name") %></td>

                        <td><%= rs.getString("email") %></td>

                        <td><%= rs.getString("city") %></td>

                        <td>

                            <a href="update.jsp?id=<%= rs.getInt("id") %>">Update</a> |
```

```

        <a href="studentservlet?action=delete&id=<%= rs.getInt("id") %>">Delete</a>
    </td>
</tr>
<%
    }
    con.close();
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
</table>
</body>
</html>

```

File name: insert.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>register Page</title>
    </head>
    <body>
        <h2>Add Student</h2>
        <form action="studentservlet" method="post">
            <input type="hidden" name="action" value="insert">
            ID: <input type="number" name="id" required><br><br>
            Name: <input type="text" name="name" required><br><br>
            Email: <input type="email" name="email" required><br><br>

```

```
City: <input type="text" name="city" required><br><br>
<input type="submit" value="Add Student">

</form>

</body>

</html>
```

File name: update.jsp

```
<%@ page contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<%@ page import="java.sql.*" %>
<%

    String idStr = request.getParameter("id");

    int id = 0;

    String name = "", email = "", city = "";

    if (idStr != null && !idStr.isEmpty()) {

        id = Integer.parseInt(idStr);

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/ajava?zeroDateTimeBehavior=CONVERT_TO_NULL",
                "root", "root");

            PreparedStatement ps = con.prepareStatement("SELECT * FROM asg3_q3 WHERE id =
?");

            ps.setInt(1, id);

            ResultSet rs = ps.executeQuery();

            if (rs.next()) {

                name = rs.getString("name");
```

```

        email = rs.getString("email");
        city = rs.getString("city");
    } else {
        out.println("<h3 style='color:red;'>Student not found.</h3>");
        return;
    }

    con.close();
} catch (Exception e) {
    out.println("<h3 style='color:red;'>Error: " + e.getMessage() + "</h3>");
    return;
}
} else {
    out.println("<h3 style='color:red;'>Error: No ID provided in the request.</h3>");
    return;
}
%>

```

```

<html>
<head>
    <title>Update Student</title>
</head>
<body>
    <h2>Update Student Record</h2>
    <form action="studentservlet" method="post">
        <input type="hidden" name="id" value="<%= id %>" />
        Name: <input type="text" name="name" value="<%= name %>" /><br/>
        Email: <input type="text" name="email" value="<%= email %>" /><br/>
        City: <input type="text" name="city" value="<%= city %>" /><br/>
    </form>

```

```
        <input type="submit" name="action" value="update"/>
    </form>
</body>
</html>
```

File name: studentservlet.java

```
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.*;
import java.sql.*;

public class studentservlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String action = request.getParameter("action");

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/ajava?zeroDateTimeBehavior=C
ONVERT_TO_NULL", "root", "root");

            if ("insert".equals(action)) {

                int id = Integer.parseInt(request.getParameter("id"));

                String name = request.getParameter("name");

                String email = request.getParameter("email");

                String city = request.getParameter("city");
```



```
        PreparedStatement ps = con.prepareStatement("INSERT INTO asg3_q3 VALUES (?,  
?, ?, ?)");
```

```
        ps.setInt(1, id);  
        ps.setString(2, name);  
        ps.setString(3, email);  
        ps.setString(4, city);  
        ps.executeUpdate();
```

```
    }
```

```
    if ("update".equals(action)) {  
        int id = Integer.parseInt(request.getParameter("id"));  
        String name = request.getParameter("name");  
        String email = request.getParameter("email");  
        String city = request.getParameter("city");
```

```
        PreparedStatement ps = con.prepareStatement("UPDATE asg3_q3 SET name=?,  
email=?, city=? WHERE id=?");
```

```
        ps.setString(1, name);  
        ps.setString(2, email);  
        ps.setString(3, city);  
        ps.setInt(4, id);  
        ps.executeUpdate();
```

```
    }
```

```
        con.close();  
    } catch (Exception e) {  
        e.printStackTrace();  
    }
```

```

        response.sendRedirect("index.jsp");
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String action = request.getParameter("action");

        if ("delete".equals(action)) {
            try {
                int id = Integer.parseInt(request.getParameter("id"));

                Class.forName("com.mysql.cj.jdbc.Driver");

                Connection con =
                DriverManager.getConnection("jdbc:mysql://localhost:3306/ajava?zeroDateTimeBehavior=C
                ONVERT_TO_NULL", "root", "root");

                PreparedStatement ps = con.prepareStatement("DELETE FROM asg3_q3 WHERE id
                = ?");

                ps.setInt(1, id);

                ps.executeUpdate();

                con.close();

            } catch (Exception e) {
                e.printStackTrace();
            }
        }

        response.sendRedirect("index.jsp");
    }
}

```

All Students

[Add New Student](#)

ID	Name	Email	City	Actions
3098	Uday	uday23@gmail.com	Ahmedabad	Update Delete

Add Student

ID:

Name:

Email:

City:

Update Student Record

Name:

Email:

City:

Q4)

File name: visitCounter.jsp

```
<%-- visitCounter.jsp --%>
```

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <title>Visit Counter</title>
```

```
</head>
```

```
<body>
```

<h1>Visit Counter</h1>

<%

// Get visit count from request attribute

Integer visitCount = (Integer) request.getAttribute("visitCount");

// If visitCount is null (direct access to JSP), handle cookie logic here

if (visitCount == null) {

 // Initialize count

 visitCount = 1;

 // Get cookies

 Cookie[] cookies = request.getCookies();

 // Check for existing cookie

 if (cookies != null) {

 for (Cookie cookie : cookies) {

 if (cookie.getName().equals("visitCount")) {

 visitCount = Integer.parseInt(cookie.getValue());

 visitCount++; // Increment count

 break;

 }

 }

 }

 // Create/update cookie

 Cookie visitCookie = new Cookie("visitCount", String.valueOf(visitCount));

 visitCookie.setMaxAge(1 * 24 * 60 * 60); // 1 days

 response.addCookie(visitCookie);

```

    }

    // Display appropriate message
    if (visitCount == 1) {
        %>
        <p>This is your first visit. Welcome!</p>
    <% } else { %>
        <p>You have visited this site <%= visitCount %> times.</p>
    <% } %>

    <p><a href="<%= request.getRequestURI() %>">Refresh the page</a></p>
</body>
</html>

```

File name: VisitCounterServlet.java

```

// VisitCounterServlet.java

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;
import java.io.IOException;

import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/counter")

public class VisitCounterServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

```

```
protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    // Get all cookies
    Cookie[] cookies = request.getCookies();

    // Initialize visit count
    int visitCount = 1;
    boolean cookieFound = false;

    // Check if visit cookie exists
    if (cookies != null) {
        for (Cookie cookie : cookies) {
            if (cookie.getName().equals("visitCount")) {
                // Get current count and increment
                visitCount = Integer.parseInt(cookie.getValue());
                visitCount++; // Increment the count
                cookieFound = true;
                break;
            }
        }
    }

    // Create or update cookie with new count
    Cookie visitCookie = new Cookie("visitCount", String.valueOf(visitCount));

    // Set cookie expiration to 30 days (in seconds)
    visitCookie.setMaxAge(1 * 24 * 60 * 60);
```

```
// Add cookie to response
response.addCookie(visitCookie);

// Set the visit count as a request attribute
request.setAttribute("visitCount", visitCount);

// Forward to the JSP
request.getRequestDispatcher("/WEB-INF/visitCounter.jsp").forward(request,
response);
}
}
```

Visit Counter

You have visited this site 5 times.

[Refresh the page](#)

Q5)

File name: login.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!-- login.jsp -->
<html>
<head><title>Login</title></head>
<body>

    <form action="loginservlet" method="post">
        <label for="username">Username:</label>
```

```

        <input type="text" id="username" name="username" required>
        <input type="submit" value="Login">
    </form>
</body>
</html>

```

File name: welcome.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!-- welcome.jsp -->
<%--<%@ page import="javax.servlet.http.Cookie" %>--%>
<%@ page import="java.net.URLDecoder" %>
<%

    String username = null;
    Cookie[] cookies = request.getCookies();
    if (cookies != null) {
        for (Cookie cookie : cookies) {
            if ("username".equals(cookie.getName())) {
                username = URLDecoder.decode(cookie.getValue(), "UTF-8");
                break;
            }
        }
    }
} %> <!DOCTYPE
html>    <html>

<head>

    <title>Welcome Page</title>

</head>

<body>

```



```

<%
    if (username != null) {
%>
        <h1>Welcome back, <%= username %>!</h1>
    <%
        } else {
%>
        <h1>Welcome, Guest!</h1>
        <p><a href="login.jsp">Login</a></p>
    <%
        }
%>
</body>
</html>

```

File name: loginservlet.java

```

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change
 this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this
 template
 */

```

```

import java.io.IOException;
import java.io.PrintWriter;
import java.net.URLEncoder;
import java.nio.charset.StandardCharsets;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;

```

```
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

public class loginservlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
        // Set response content type
        response.setContentType("text/html;charset=UTF-8");

        // Retrieve the 'username' parameter from the request
        String username = request.getParameter("username");

        try (PrintWriter out = response.getWriter()) {
            if (username != null && !username.trim().isEmpty()) {
                // Encode the username to ensure it's safe for cookie storage
                String encodedUsername = URLEncoder.encode(username,
StandardCharsets.UTF_8.toString());

                // Create a cookie with the encoded username
                Cookie userCookie = new Cookie("username", encodedUsername);
                userCookie.setMaxAge(60 * 60 * 24 * 7); // Cookie valid for 7 days
                response.addCookie(userCookie);

                // Display a welcome message
                out.println("<!DOCTYPE html>");
                out.println("<html>");
                out.println("<head>");
                out.println("<title>Welcome</title>");
```

```

        out.println("</head>");
        out.println("<body>");
        out.println("<h1>Welcome, " + username + "!</h1>");
        out.println("<p><a href=\"welcome.jsp\">Go to Welcome Page</a></p>");
        out.println("</body>");
        out.println("</html>");
    } else {
        // If username is invalid, prompt the user to enter it
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Login Error</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h1>Error: Username cannot be empty.</h1>");
        out.println("<p><a href=\"login.jsp\">Try Again</a></p>");
        out.println("</body>");
        out.println("</html>");
    }
}
}
}

```

@Override

```

protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    // Redirect GET requests to the login page
    response.sendRedirect("login.jsp");
}

```

@Override

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    // Process POST requests
    processRequest(request, response);
}
```

@Override

```
public String getServletInfo() {
    return "Login Servlet that stores username in a cookie";
}
}
```

Username:

Welcome, Uday!

[Go to Welcome Page](#)

Welcome back, Uday!

Q6)

File name: displayBooks.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<html>
```

```

<head>

    <title>Library Books</title>

</head>

<body>

    <h2>Book List from Database</h2>

    <table border="1" cellpadding="10">

        <tr>

            <th>Title</th>

            <th>Author</th>

            <th>Price (₹)</th>

        </tr>

<%

    String url = "jdbc:mysql://localhost:3306/ajava?useSSL=false";

    String user = "root";

    String password = "root";

    try {

        Class.forName("com.mysql.cj.jdbc.Driver");

        Connection conn = DriverManager.getConnection(url, user, password);

        Statement stmt = conn.createStatement();

        ResultSet rs = stmt.executeQuery("SELECT * FROM asg3_q6");

        while (rs.next()) {

%>

        <tr>

            <td><%= rs.getString("title") %></td>

            <td><%= rs.getString("author") %></td>

            <td><%= rs.getDouble("price") %></td>

        </tr>

```

```

<%
    }

    rs.close();
    stmt.close();
    conn.close();
} catch (Exception e) {
    out.println("<tr><td colspan='3'>Error: " + e.getMessage() + "</td></tr>");
}
%>

</table>

</body>

</html>

```

File name: web\WEB-INF\web.xml

```

<?xml version="1.0" encoding="UTF-8"?>

<web-app version="6.1" xmlns="https://jakarta.ee/xml/ns/jakartaee"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
https://jakarta.ee/xml/ns/jakartaee/web-app_6_1.xsd">

    <servlet>

        <servlet-name>books</servlet-name>

        <servlet-class>books</servlet-class>

    </servlet>

    <servlet>

        <servlet-name>ParseServlet</servlet-name>

        <servlet-class>ParseServlet</servlet-class>

    </servlet>

    <servlet-mapping>

        <servlet-name>books</servlet-name>

```

```
<url-pattern>/books</url-pattern>
</servlet-mapping>
<servlet-mapping>
    <servlet-name>ParseServlet</servlet-name>
    <url-pattern>/ParseServlet</url-pattern>
</servlet-mapping>
</web-app>
```

File name: web\WEB-INF\books.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<!--
```

Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license

Click nbfs://nbhost/SystemFileSystem/Templates/XML/XMLDocument.xml to edit this template

```
-->
```

```
<library>
    <book>
        <title>The Alchemist</title>
        <author>Paulo Coelho</author>
        <price>299.99</price>
    </book>
    <book>
        <title>1984</title>
        <author>George Orwell</author>
        <price>349.50</price>
    </book>
    <book>
```

```
<title>To Kill a Mockingbird</title>
<author>Harper Lee</author>
<price>275.00</price>
</book>
<book>
  <title>Pride and Prejudice</title>
  <author>Jane Austen</author>
  <price>310.75</price>
</book>
<book>
  <title>The Great Gatsby</title>
  <author>F. Scott Fitzgerald</author>
  <price>199.99</price>
</book>
<book>
  <title>Harry Potter and the Sorcerer's Stone</title>
  <author>J.K. Rowling</author>
  <price>450.00</price>
</book>
<book>
  <title>The Hobbit</title>
  <author>J.R.R. Tolkien</author>
  <price>399.00</price>
</book>
<book>
  <title>The Catcher in the Rye</title>
  <author>J.D. Salinger</author>
  <price>280.25</price>
</book>
```



```
<book>
  <title>Life of Pi</title>
  <author>Yann Martel</author>
  <price>330.40</price>
</book>
<book>
  <title>Wings of Fire</title>
  <author>A.P.J. Abdul Kalam</author>
  <price>240.00</price>
</book>
</library>
```

File name: ParseServlet.java

```
import java.io.File;
import java.sql.*;
import javax.xml.parsers.DocumentBuilder;
import javax.xml.parsers.DocumentBuilderFactory;
import org.w3c.dom.*;

public class ParseServlet {
    public static void main(String[] args) {
        try {
            // Load XML file

            File xmlFile = new File("C:/Users/uday/Ajava/assignment3/asg3_que6/web/WEB-INF/books.xml");

            // DOM parser
            DocumentBuilderFactory factory = DocumentBuilderFactory.newInstance();
            DocumentBuilder builder = factory.newDocumentBuilder();
```

```
Document doc = builder.parse(xmlFile);

doc.getDocumentElement().normalize();

NodeList nodeList = doc.getElementsByTagName("books");


// MySQL connection
String url = "jdbc:mysql://localhost:3306/ajava?useSSL=false";
String user = "root";
String password = "root";
Connection conn = DriverManager.getConnection(url, user, password);

String insertQuery = "INSERT INTO asg3_q6 (title, author, price) VALUES (?, ?, ?)";
PreparedStatement ps = conn.prepareStatement(insertQuery);

for (int i = 0; i < nodeList.getLength(); i++) {
    Element book = (Element) nodeList.item(i);
    String title = book.getElementsByTagName("title").item(0).getTextContent();
    String author = book.getElementsByTagName("author").item(0).getTextContent();
    String price = book.getElementsByTagName("price").item(0).getTextContent();

    ps.setString(1, title);
    ps.setString(2, author);
    ps.setDouble(3, Double.parseDouble(price));
    ps.executeUpdate();
}

System.out.println("Books inserted successfully!");

ps.close();
```

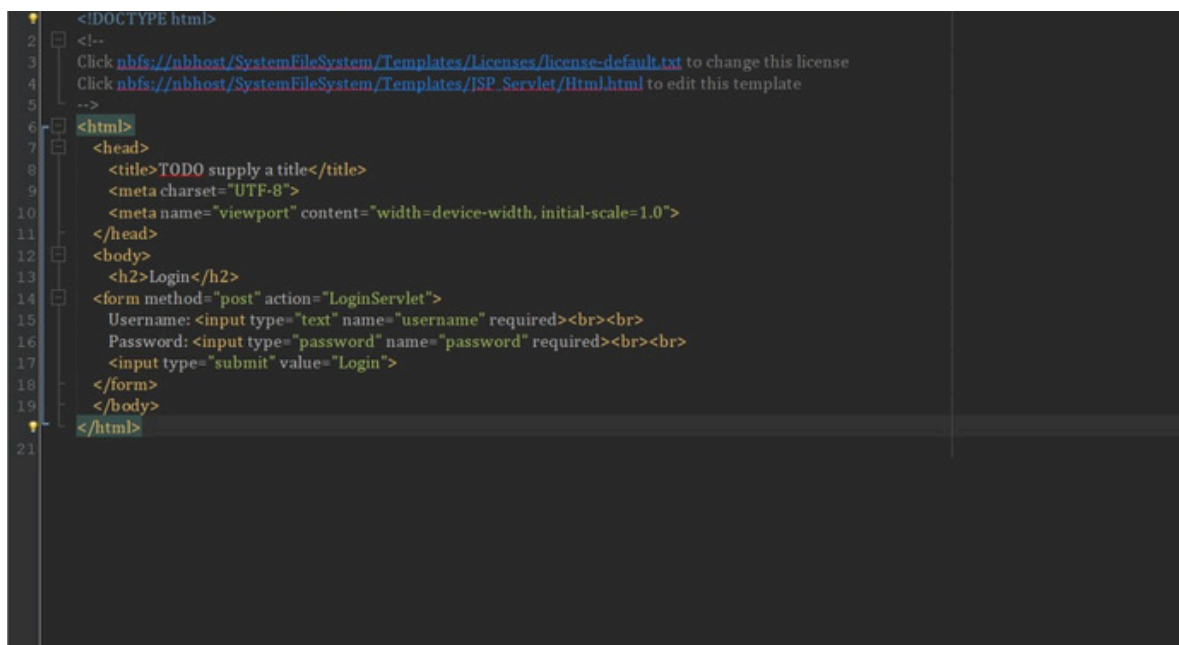
```

        conn.close();
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}
}

```

Q7)

File name: login.html



```

1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5 -->
6 <html>
7 <head>
8 <title>TODO supply a title</title>
9 <meta charset="UTF-8">
10 <meta name="viewport" content="width=device-width, initial-scale=1.0">
11 </head>
12 <body>
13 <h2>Login</h2>
14 <form method="post" action="LoginServlet">
15 Username: <input type="text" name="username" required><br><br>
16 Password: <input type="password" name="password" required><br><br>
17 <input type="submit" value="Login">
18 </form>
19 </body>
20 </html>
21

```

File name: web\WEB-INF\classes\users.xml

```

<users>

  <user>

    <username>admin</username>

    <password>admin123</password>

  </user>

  <user>

```

```
<username>john</username>
<password>john123</password>

</user>

</users>
```

File name: LoginServlet.java

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change
 this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/JSP_Servlet/Servlet.java to edit this
 template
 */
```

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
```

```
/**
 *
 * @author Uday
 */
```

```
import java.io.*;
import javax.xml.parsers.*;
import org.w3c.dom.*;
```

```
public class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
```

```

// Retrieve input from the form
String username = request.getParameter("username");
String password = request.getParameter("password");

// Load and parse the XML file
InputStream xmlFile = getServletContext().getResourceAsStream("/WEB-INF/classes/users.xml");

if (xmlFile == null) {
    response.getWriter().println("User data file not found.");
    return;
}

try {
    // Parse the XML file
    DocumentBuilderFactory factory = DocumentBuilderFactory.newInstance();
    DocumentBuilder builder = factory.newDocumentBuilder();
    Document doc = builder.parse(xmlFile);
    doc.getDocumentElement().normalize();

    // Retrieve all user elements
    NodeList userList = doc.getElementsByTagName("user");
    boolean isAuthenticated = false;

    // Iterate through user elements to find a match
    for (int i = 0; i < userList.getLength(); i++) {
        Node userNode = userList.item(i);
        if (userNode.getNodeType() == Node.ELEMENT_NODE) {
            Element userElement = (Element) userNode;
            String xmlUsername =
userElement.getElementsByTagName("username").item(0).getTextContent();

```

```
String xmlPassword =  
userElement.getElementsByTagName("password").item(0).getTextContent();
```

```
    if (username.equals(xmlUsername) && password.equals(xmlPassword)) {  
        isAuthenticated = true;  
        break;  
    }  
}  
}
```

```
// Respond based on authentication result  
response.setContentType("text/html");  
PrintWriter out = response.getWriter();  
if (isAuthenticated) {  
    out.println("<h2>Login Successful</h2>");  
    // Redirect or forward to a protected resource or dashboard  
} else {  
    out.println("<h2>Invalid Username or Password</h2>");  
    // Optionally, redirect back to the login page  
}  
} catch (Exception e) {  
    throw new ServletException("Error processing login", e);  
}  
}  
}
```

Login

Username:

Password:

Login

Invalid Username or Password

Login

Username:

Password:

Login

Login Successful